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Do they work? Exploring possible potentials of neighbourhood Telecommuting centres in supporting sustainable travel

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ABSTRACT

Digitally enabled neighbourhood telecommuting centres (NTCs) in residential areas may have the potential to reduce the environmental burdens of transport by shortening work trips and enabling modal shifts. This paper presents the results of a Living Lab where 67 participants were given access to an NTC.

Through this Living Lab, this study identifies several conditions required for an NTC to substantially reduce commuting and overall have a positive impact on sustainable travel. The results indicate that while a small group of participants who lived very close to the NTC made significant lifestyle changes and adopted more sustainable travel practices, the overall changes in the sustainability aspects of travel for most participants were minimal.

The majority of the participants merely exchanged a day spent working from home for a day at the NTC, as they were only allowed to be absent from the employer's office one day per week. Further, some participants found it difficult to work remotely due to organisational roles and workplace norms. Another factor that limited the sustainability effect of the NTC was that most participants normally commuted by train, but in a few cases travelled to the NTC by car.

With the arrival of the COVID-19 pandemic, most participants were required to work exclusively from home, which proved to have both positive and negative effects on the participants' everyday lives and well-being.

The participants reported a radical shift in attitudes towards remote working during this period, which they believed could lead to remote working becoming more common after the pandemic. In this case, NTCs could possibly play an important role in enabling this shift, by remedying some of the identified drawbacks of working from home. This could in turn enable a larger reduction in commuting. In areas where more people commute by car, sustainability effects would likely be greater.

1. Introduction

Transportation is currently responsible for a large share of greenhouse gas emissions. In the European Union, 27% of greenhouse gas emissions came from the transport sector in 2017, of which cars accounted for 60% (European Parliament, 2019). According to the UNFCCC (2019), to meet reduction targets of CO2 emission by 2030, national, subnational, and local policies need to support changes in travel behaviour that will result in a shift to sustainable modes of transport, and transition to a more holistic urban development planning that prioritises multimodal integration. In recent years, innovations in Information and Communication Technologies (ICT) and digital platforms have enabled the development of combined mobility and accessibility services, such as Mobility as a Service (MaaS). MaaS allows different providers of mobility services, such as public transport, public transport on demand, and rental bicycles, to offer their services through a common digital platform (Pangbourne et al., 2018; Jittrapirom et al., 2017; Maeng & Nedovic-Budic, 2010; Wong et al., 2020). By including services such as food delivery services, remote meetings, and Neighbourhood Telecommuting Centres (NTC), these combined services could replace or shorten the travel and could hold promising potential in reducing travel-related environmental impacts in the long term (Mokhtarian, 1990; Jittrapirom et al., 2020; Ringenson and Kramers, 2021; Kramers et al., 2015; Sopjani et al., 2018).

In this paper, we explore the concept of NTCs, which can be described as office spaces shared by employees of two or more

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employers to enable remote working. When combined with sustainable modes of transport, NTCs could hold the potential to reduce long-distance commuting as well as avoid the disadvantages of working from home, such as lacking ergonomics and social isolation (Mokhtarian, 1990; Buffer, 2021; Vaddadi et al., 2020) by providing a "third place" to enable involvement with others outside the family and work sphere (Oldenburg & Brissett, 1982). Empirical studies on home working show that although it may reduce commuter travel, it may also lead to additional trips and urban sprawl (Moeckel, 2017). It can for instance lead to an increase in leisure and recreational travel (Marvin 1994)) such as (i) a desire to travel to participate in activities identified via ICT, (ii) the time saved being used to make other trips and (iii) dispersed residential and employment locations that result in longer trips (Mokhtarian, 1990)].

Understanding the potentials of a new service or technology is not straightforward, as how it is appropriated and used can rarely be understood before it is in use (Henderson and Kyng, 1991). From studies on technology innovation failures, Spaargaren (2011) concludes that products, technologies, or infrastructures for sustainability purposes must be designed with knowledge of user practices. An NTC's direct effects on commuting may be offset by unexpected changes to other travel practices, or the uptake be limited by seemingly unrelated factors, and there is often a long process of adaptation before such effects can be known. Real-life experimentation methods such as Living Labs make it possible to explore the potentials of innovations (Evans and Karvonen, 2010).

This article presents findings from a Designerly Living Lab (DLL) (Sjöman and Hesselgren, 2020). Compared to other types of Living Labs, DLLs have a more explorative approach, where the participants are exposed to future scenarios, staged in the context of their everyday lives in the form of "design interventions". Significantly, these interventions are not seen as product or service concepts to be evaluated and refined, but as "learning devices" (ibid.). Through interviews, tracking, and observation over time, it becomes possible to understand how the interventions are perceived by the participants, and how their every day lives may be affected. The research presented in this article is based on the DLL "*Work Closer, Travel Smarter*", in which an online platform, including an app, was set up, combining mobility services with non-travel options in the form of an NTC, to explore its potential to support more sustainable travel in connection with work.

In this paper, we use the ASI framework to analyse the findings in order to clarify the potential of NTCs in combination with e-bicycle sharing to support sustainable travel, especially in connection with commuting.

The Avoid-Shift-Improve (ASI) framework has been suggested as a useful tool to understand the transition to sustainable transport solutions (United Nations, 2016; Dalkmann and Brannigan, 2007). An ASI-based approach suggests that policy and planning to seek to:

Avoid or reduce unnecessary travel where appropriate (United Nations, 2016). This means that travel does not take place or that distances are shortened, for example with the help of teleworking or compact, multimodal communities (United Nations, 2016; Dalkmann and Brannigan 2007; Lah, 2017).
Shift from private-owned, low-occupancy cars to more efficient, low-carbon modes, such as active modes, public transport, and ride-sharing (Dalkmann and Brannigan, 2007; United Nations, 2016; Lah, 2017).
Improve the environmental performance of transport systems and vehicles (Tiwari et al., 2011; United Nations, 2016).

However, as a physical workplace cannot play a role in *improving* transport, we explore the potential of an NTC in helping to *avoid* transport by allowing fewer or shorter commuter trips, as well as related *shifts* in travel modes or travel times.

In a recent study (Ringenson & Kramers, 2021) it was noted that the connection between MaaS and "Avoid" in the ASI framework is almost non-existent in literature, and in interviews with public authorities, it was shown that the representatives found it difficult to connect non-travel services with mobility services. As will be presented in the discussion, in the present study, this disconnect between avoiding travel and transportation perspectives has been further explored.

After the DLL had been in place for about a year, conditions for office work changed drastically due to the SARS-CoV-2 pandemic (COVID-19). On March 16th, 2020, the Swedish Public Health Department issued recommendations for people to work from home as much as possible to prevent the spread of COVID-19 (The Public Health Agency of Sweden, 2020). This meant that practically all of the DLL participants started working from home. While this initially led to lower use of the NTC, it also changed the conditions for remote working overall.

1.1. Aim and research questions

This article aims to explore the potential of NTCs to contribute to more sustainable travel. The study was guided by the following research questions:

Did participants *avoid* transport by making fewer or shorter trips when they were given access to the NTC? if so, how and under which conditions?
Did participants *shift* travel modes or travel times for trips when they were given access to the NTC? If so, how and under which conditions? And what were the participants' motivations and which effects did these shifts have?

How can NTCs support *avoiding* and *shifting* in the future? which conditions need to be changed for NTCs to better support these?

The paper is divided into five sections. In section two, we discuss the research methods, including data collection and analysis. In section three, we present the results of the analysis, followed by a discussion of our findings in section four. These results regard both travel itself and the underlying conditions related to work. In section five, we provide our concluding takeaways.

2. Research METHODS

2.1. The DLL work closer, travel smarter

This article presents results from the DLL *Work Closer, Travel Smarter*, which was centred around an NTC that was set up in a largely residential villa suburb area in the municipality of Botkyrka, 20 km south of Stockholm. The NTC was set up in a neighbourhood centre, close to the local library, supermarkets, and similar services, and in close proximity to a commuter train station.

It may be noted that the commuter train has a high passenger capacity, is well-used for commuting, gets crowded during rush hours, and sometimes experiences delays (Trafikverket, 2020). During rush hours, the train leaves towards the Stockholm city centre every four to eight minutes (SL, 2021). Several buses also leave from the station: Mostly local bus lines, and two commuter lines that go into the Stockholm city centre. The area is located close to a motorway that is often congested during rush hours.

The 170 sq meters NTC offered a fully-equipped activity-based office space with 14 bookable desks, three telephone booths, a conference room, a kitchen, and a lounge area. Two electric bicycles and one electric box bicycle were also bookable through an app and parked outside the NTC. As the NTC was intended to function as an accessibility service to support shorter commutes, the recruitment of participants focused on commuters who lived in close vicinity to the NTC. Participants were recruited in two rounds. Group 1 (32 people) was recruited via local Facebook groups and Facebook ads targeting commuters in the neighbourhood, as well as by word of mouth within partner

organisations of the research programme.

Participants in Group 1 worked at various workplaces and often commuted long distances. In this group, most people lived very close to the NTC, i.e. less than a 30-minute walk away. Group 2 (35 people) was recruited through a large partner organisation associated with the research programme. This round of recruitment happened later, as participation in the DLL had to be cleared at several levels of the organisation. The organisation's human resources department then identified employees who lived within a 5 km radius of the NTC and offered these employees the opportunity to work at the facility as a complement to their normal office attendance. In total, 67 people registered as participants in the Living Lab, and 52 ended up actively using the NTC.

2.1.1. User and usage statistics before working at the NTC

Initial data about the respondents was collected from an application form they completed in order to participate in the Living Lab, and covers information about their 1) normal work location, 2) normal travel mode to work, 3) normal travel mode outside of work, and 4) average commuting time. Additionally, a survey was administered to gather information about demographics, travel habits, and various characteristics of respondents' current working life (Table 1).

More than 50% of the respondents worked in Kista or in Solna, north of Stockholm. Ten respondents worked in the city centre of Stockholm; two worked in Södertälje, a municipality south of Tullinge; two respondents worked mainly from home; and one worked in another city four hours away by train. The large employer in Kista previously had office spaces in southern Stockholm, much closer to the location of the NTC, and several of the participants had settled in Tullinge at that time.

Most of the respondents normally travelled by commuter train to work, and owned a fixed-price commuter card. Only a limited number travelled by car. Those who lived far away from the commuter train also travelled by bus, bicycle, or car to the train station. Two respondents mainly commuted by bike.

A majority of the respondents indicated that they used their own car for most non-work-related travel.

The average commuting time for all respondents was approximately 60 min in one direction. The time varied from 40 up to 80 min in one direction.

Most respondents worked regularly at their employer's office and from home one to three days a month. After March 2020, due to the COVID-19 outbreak, most participants worked almost exclusively from home, and only a small number used the NTC. See section 3.

2.2. Data collection

During participation in the DLL, the participants took part in in-person interviews, phone call interviews, and three-week time-use diaries. Use of the NTC and the bicycles was tracked by logging data from the booking platform. During the course of the DLL, it became clear that most participants found it challenging to work more than one day per week at the NTC (see section 3). Therefore, to explore a possible future where societal structures might better support remote working, a mixed group of 10 "challenge takers" were asked to experiment with working three to four days per week at the NTC for a period of three weeks. During this time, they filled in an additional diary and were subsequently interviewed about their experiences.

2.2.1. Interviews

To capture qualitative data on the participants' thoughts, feelings, and reflections, as well as more detailed accounts of their work life, needs, and travel habits, interviews were conducted. The interviews were semi-structured to allow for a combination of structure and flexibility (Edwards & Holland, 2013). Four main sets of interviews were carried out:

Table 1
Participant characteristics based on demographics, travel habits, and current working life.

Variable		Count (n= 52)
Age	25-35	2
	35-45	10
	45-55	19
	55-65	21
Gender	Male	31
	Female	21
Household Size	1	1
	2	17
	3	7
	4	17
	5 or more	10
Number of respondents with children in the household	None	18
	One	6
	Two	18
	Three	9
	Four	1
Number of respondents with children in the age group of	0-10	17
	10-20	15
	20 and above	6
	No data	14
Commuting mode preferences *(represents the main travel mode used in a commuting trip)*	Car	11
	Public Transport	36
	Bike	3
	Foot	2
	Other modes	0
Leisure travel mode preferences	Car	32
	Public Transport	8
	Bike	2
	Foot	9
	Other modes	1
Commuting distance *(one way from home location to work location)*	Less than 15 kms	1
	15 kms- 20 kms	3
	20 kms- 25 kms	11
	More than 25 kms	37
Type of current employer's office	Activity- based open workspaces	7
	Open workspaces with fixed desks	27
	Office shared with other employees	3
	Private office	3
	No data	12
Days/week at employer's office	5	16
	4	17
	3	6
	1	1
	No data	12
Days/week working from home	1-3 days/ month	24
	3-4 days/ week	3
	1-2 days/ week	10
	0 days/ week	3
	No data	12
Days/week expectations of working from the NTC	1-3 days/ month	9
	3-4 days/ week	4
	1-2 days/ week	26

(*continued on next page*)

Table 1 (*continued*)

Variable		Count (n= 52)
	0 days/week	1
	No data	12
Satisfaction with current employer's office	Satisfied	22
	Somewhat satisfied	10
	Not satisfied	7
	No data	12

Initial interviews at the start of the DLL, with participants from several different workplaces (Group 1). All 15 participants active at that time were interviewed. Questions were structured around the themes of expectations for the study, everyday life, travelling habits, and work life. The interviews were performed in person, usually at the site of the NTC, and lasted between 45 and 70 min.

Initial interviews with the second group of 30 participants, who all joined from the same workplace (Group 2). The structure of these interviews was nearly identical to that of Group 1 with a little more emphasis on everyday life.

Interviews with 8 participants (1 participant from Group 1 and 7 participants from Group 2) who took part in a "challenge" to work more days than usual at the NTC. A challenge diary was completed for three weeks, after which a follow-up phone interview was conducted with each of the 8 participants. These interviews lasted around 30–40 min, the main topics being problems, needs, and solutions related to spending more time at the NTC, quality of life, and effects on other daily routines.

Shorter phone interviews (10–15 min) with 40 participants (11 participants from Group 1 and 29 participants from Group 2) in spring 2020, five to six weeks after the Swedish Public Health Agency issued recommendations to work from home due to the COVID-19 outbreak. Questions mainly concerned experiences of working (almost exclusively) remotely from home.

2.2.2. Three-week travel diaries

The 52 participants, active at this time, received a three-week time-use diary in which to record their travel- and work-related activities for a period of three weeks, on weekdays, and at weekends. The diary was designed to indicate four types of activities (work, travel, everyday chores, and leisure), and time spent travelling via different transport modes during each diary day. Out of the 52 participants, 27 participants (including 8 "challenge takers") regularly filled in the diary and handed it in by the end of the three-week period.

2.3. Data analysis

2.3.1. Qualitative analysis

The interview data was analysed in accordance with the qualitative content analysis method, as described by Schreier (2014). Qualitative content analysis suggests a coding strategy that departs from a theoretical framework and is then adjusted and complemented by data-driven codes. It is especially useful for focusing on aspects of meaning in the material that relate to the overall research question. This method was considered suitable for the data in this article, as this data is fairly heterogeneous. Coding was performed using the MaxQDA software.

When conducting the analysis, we started out with a theoretically derived coding framework built around the Avoid and Shift components of ASI. After two coding rounds, we added new categories that were identified as needed to explain and contextualize how travel was or was not avoided. Notably, these codes included working conditions and well-being related aspects and experienced differences between remote working and working at the employer's office. The last distinction was important to make, as the exploration of the data showed that the participants often spoke more broadly about going to their employer's office or working elsewhere, rather than differentiating between working from home and working from the NTC. We also observed a need to separate problems that were inherent in remote working from problems that could possibly be solved in the future with an improved NTC service offering.

2.3.2. Quantitative analysis

The content of the 27 travel diaries was used to understand the time spent on commute- and work-related activities over a period of three weeks during the months of September and October 2019. A total of 416 weekdays were recorded, out of which 227 days were recorded to be employer's office days, followed by 76 days at the NTC and 49 days working from home (Table 2).

We compared average time spent on 'travel', 'work', and 'virtual meetings' as well as average time spent using different travel modes on weekdays, when participants worked from the employer's office, from the NTC, or from home. We did not consider the days participants worked from other locations or alternated between two or more locations, as this accounted for a very small proportion of the time.

3. Limitations

The interviews used for the data collection in this article were largely not transcribed. Instead, the researchers took extensive notes, and unclear or particularly important quotes were checked against recordings. Due to the large time frame of the study (two years at the time of writing), some participants joined at a later date than others, while others left before the study ended. Since participants joined the DLL in different ways, with some being recruited through their workplace and others through advertising, the benchmark data is not fully reliable. As some participants sometimes failed to use the software to book a place at the NTC, manual, retroactive corrections of the booking data were made, based on check-up interviews. The time-use diary did not cover all the participants, as only 27 out of 52 participants properly completed and handed in the diaries. However, this represented the large majority of the participants who actively used the NTC at the time. For the analysis of the time-use diaries, we only considered the weekday data, as there were many missing entries in the weekend data. However, since the main focus of this study was to understand how the participants worked and commuted, we had sufficient observations (411 weekdays) to carry forward the analysis.

4. Results

Our study indicates that the NTC became an alternative to going into the office, as well as an alternative to working from home. Which of these was dominant seems to differ between the participants. Five of the participants clearly stated that they used the NTC instead of working from home, for example:

"I*t is instead of working at home, which would have been the alternative otherwise.*"

Other statements indicate that the NTC was largely considered to be an extension of and an alternative to the employer's office:

"Who doesn't dream of working closer to home?".

4.1. Avoiding travel

When we initially asked the participants about their motivations for participating in the study, their answers were mainly related to avoiding long commutes (16 participants). This was generally either explicitly formulated as *"save/saving time"* or *"to reduce travel time"*, or expressed more indirectly in comments about how tiring the long commutes were.

Table 2
Number of days recorded in the travel diaries for each workplace.

NTC	Employer's office	Home office	Meetings outside of workplace	Combined workplaces	No work	No data recorded
76 days	*227 days*	*49 days*	*5 days*	*39 days*	*8 days*	*7 days*

"I try to adjust so I can make it back in time for the children's activities. And you always have to count delays into the equation. It is very tiring and stressful with the travel time and finding time for family life."
"I experience less stress if I can go home sometime in the middle of the day and sit here [at the NTC] for a few hours and work before I have to pick up [the kids]."

The results of the travel diary hence show a significant reduction in time spent on commuting when the participants worked from the NTC when compared to the days they commuted to the employer's office (Fig. 1).

However, as previously mentioned, several of the participants exchanged working from home for working at the NTC, thereby adding a trip to the NTC in the process and thus not clearly *avoiding* travel. However, some were also able to do more remote working thanks to the NTC, especially those who were not normally allowed to work from home. One person however stated that this shift was enabled because his employer was a partner in the research programme, rather than because of the NTC in and of itself.

4.1.1. Separating work and free time

When comparing working at the NTC to working from home, participants often described being easily distracted in the home setting by e.g. family members and household chores:

"I have sat at home a little before but then it is also… you see the sink, you see the dirty socks, and then you also get a little distracted."

Another participant compared the NTC to both the home and the employer's office:

"It [the NTC] is smaller than the space at [employer's office]. You show more consideration, so you get to work undisturbed. You work in a more focused way without matters at home attracting your attention."

A better division between work time and leisure time was also a reason to go to the NTC instead of working from home:

"If you work a lot from home, you cannot distinguish between free and work time at all."
"[The NTC] is better than sitting at home, more like a real job; it supports discipline."
"It does not work to work at home. You are disturbed by the children, and there is no clear demarcation between work and leisure."

However, some participants stated that they intentionally worked from home in order to combine household chores with paid work, while others said that they preferred the home to the NTC as it required no trip at all:

"Like this morning when I worked at home, then you can turn on the washing machine."
"Broadly, I work remotely to save travel time. Everything is set up at home, I can just sit down [and start working]."

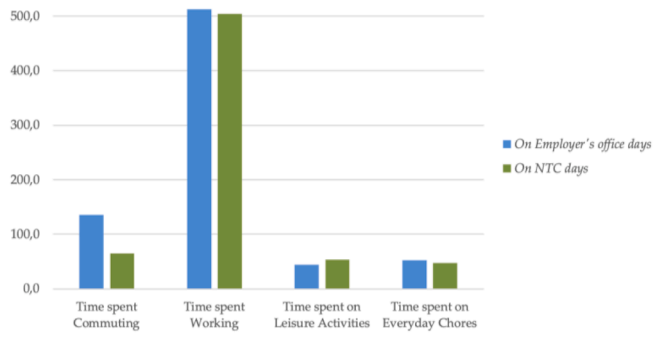


Fig. 1. A comparison of time spent (in mins) on various activities on employer's office and NTC days based on the travel diaries.

4.1.2. Localisation of the NTC and connection with public transport

As previously mentioned, the NTC was located very close to the commuter train station. Several of the participants mentioned the localisation of the NTC as an important factor. Firstly, as most participants normally went to work by commuter train, their trip to the NTC was often essentially the same as the first part of their normal commute:

"I'll just go to the [NTC] instead of the station."

Secondly, several participants used the NTC as a landing point between their office and home, or as a place to work after external meetings, when it would be too time-consuming to go back to their employer's office but they did not want to go straight home.

One participant described her time at the NTC as "a transition period" that gave her "a taste of working locally":

"So, I started looking for new jobs, which would be within cycling distance. It felt so nice to hang out with local people. Then I applied for a job [closer to home]. [After starting the new job] my quality of life has improved. Cycling to work makes me feel so much better. Previously I've been on sick leave [due to stress], commuting sitting among lots of people, stressing to get to the train. /…/ Now I go by car or bicycle. Before, I had like two lives; one life in the city, and one here. Now it's more like – it feels more safe to be closer to my children. I can go home at lunchtime if I have to pick something up. /…/ The days I worked at the [NTC], that I did not take the train, /…/ I felt much more relaxed on those days."

4.2. Shifting modes of travel

Most of the participants normally commuted by train, and to a large extent, participants who would normally walk to the station walked to the NTC as well. Those who normally drove to the employer's office also drove to the NTC, although for a shorter distance. According to the interviews, some shifts in travel modes occurred; these were mainly sustainable, but there were also instances of commuting by car instead of public transport. The stated reason for taking the car was that there was no congestion going in the direction of the NTC.

These patterns are clearly visible in the travel diaries, and Fig. 2 shows the comparison of time spent on various travel modes on employer's office and NTC days. As most participants travelled to work by PT, this is where the largest reduction is seen. On the other hand, four participants who often used the NTC reported shopping more locally on days when they used the facility, which may have reduced car use.

Two participants that shifted to commuting by car to the NTC claimed that this allowed them to combine the shorter commute with errands, which means that the added car ride may have been offset by avoiding another trip.

4.3. Conditions for the NTC to support avoiding and shifting

One important finding of the living lab was that the main barriers for NTCs to enable a reduction in travel are related to workplace policy, routines and norms. For this reason, we go into some detail in describing

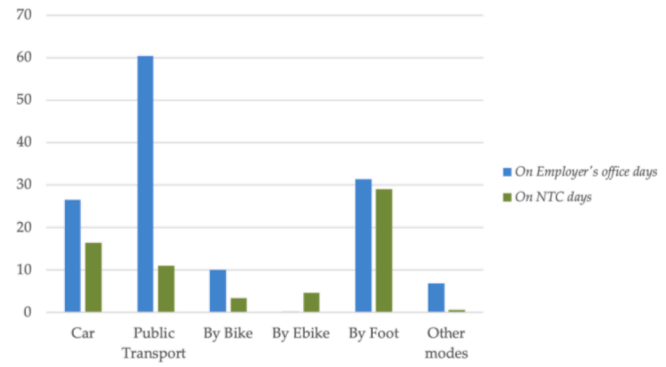


Fig. 2. A comparison of time spent (in mins) commuting in various travel modes on employer's office and NTC days based on the travel diaries.

these barriers. As previously mentioned, before the COVID-19 pandemic most of the participants were not allowed to work remotely more than one day a week, and often less.

4.3.1. Working without distractions

During the interviews, participants' reflections on working from home and from the NTC were often interwoven, as both are ways of working remotely. For many participants, being able to work without being disturbed was a reason to work from home as well as to work from the NTC. Furthermore, several participants also said that it was an advantage being unfamiliar with the other users at the NTC, because it meant there were fewer distractions:

"I get peace of mind from working at the NTC. Peace and quiet are needed sometimes."
"If I need to sit in concentration, it can be better to sit at home to avoid getting interrupted."

4.3.2. Co-worker collaboration and workplace norms

The need for close collaboration and communication with colleagues was clearly one of the most important conditions limiting the NTCs effects on travel. The two participants who used the NTC most frequently described their roles as highly independent in nature. Conversely, entering a new role that required more interaction with others was one of the two most common reasons stated for not using the NTC. Other commonly mentioned hindrances had to do with policies, regulations and norms at work, and expectations from managers. The more flexible and permissive managers would allow working from home one day per week, and for most participants, being away from the office more often was simply not allowed. One participant said that he probably would be allowed to work from home more, but that he needed to "be visible" at the office to be considered for promotions. In most cases, there did not seem to be clearly set rules, but many participants shared the understanding that regulations regarding insurance and employers' responsibility for the workplace environment were reasons for strict policies against working remotely. Some of the problems around collaboration had to do with the physical conditions for taking phone calls and conducting remote meetings from the NTC, which many participants considered inadequate. While the NTC was equipped with three phone booths, these turned out to be too few, and the sound insulation and ventilation was also deemed insufficient:

"The phone booths are just terrible. It is no good for sensitive conversations, and it's too warm."

Some of the participants would be in remote meetings for up to six hours per day, for which they stated a need for a slightly larger, private room. These participants generally ended up choosing to work from home instead of at the NTC. In contrast, the six-person meeting room was very rarely booked, since most participants did not have any direct colleagues at the NTC. The subject of computer monitors was mentioned by several of the participants. The computer monitors at the NTC were seen by some as an improvement compared to working from home, and by others as a limitation compared to their employer's office:

"[I miss] having two screens. I have three at work. It's more comfortable."

4.3.3. Decreased use of the NTC

Among the participants who quit or decreased their use of the NTC, two reasons stood out as most common: favouring working from home over working from the NTC, and entering a new role at work. Sometimes, these were combined, as when a new role involved long remote meetings and due to the inadequate phone booths at the NTC, the home was considered more comfortable. One of the participants said:

"I tried but realised that it [the NTC] did not add anything directly, compared to work from home. Home was more convenient. It did not get very social at the [NTC] either. I didn't have much use of video rooms and such. I have good conditions for working at home."

During the COVID-19 pandemic, one participant also stated:

"Now that I am allowed to work from home, it feels unnecessary to go to the [NTC] (three commuter stations away, authors' remark). *Now I have also picked up both screens and a chair from the office."*

Among those who entered new roles, the most common reason for not using the NTC was that their new roles involved more meetings and presence at the office. One of the participants explained:

"My current role is as a project manager for a part of a large IT project. /…/ My role is a lot about attending meetings but also being available to people in the project who need to ask or coordinate things. My role right now is a lot about creating a sense of security, clarity, and direction, and I feel that requires a personal presence."

Several of the participants also mentioned that it was difficult to form habits and routines around going to the NTC. Many thought that they would at some point start using the NTC more often, but most participants found it difficult to get into the habit:

"Every time you come here you become like, 'Gosh, I really have to get here [more often], because it feels like a quiet environment where you can get a lot done. But then taking the step to actually come… It's like the gym or something."
"I have intended to work more at home but have delayed it. It is difficult to break habits."

3.3.5 Changed conditions under the COVID-19 pandemic.

In March 2020, many workplaces requested their employees to work from home, when possible, to avoid the spread of COVID-19. This included the workplaces of the DLL participants. Most of the participants were of the opinion that experiences of remote working during this time would have a long-term impact, and lead to remote working becoming more common in future.

Several participants perceived a need for more NTCs to emerge if that were to be the case, to alleviate the downsides of working remotely. One participant, who had been an occasional user of the NTC, said that he found himself working less effectively but for longer hours from home, while still disliking the long commute to the main office. When asked about his thoughts on the role of NTCs after the pandemic, he answered:

"I do not have to sit at the [employer's office] all the time. This, if nothing else, has been proven. The [NTC] creates a bit more routine than working from home; you come to work and then you go home."

When asked about the working life after the pandemic and the role of NTCs, this participant said that he anticipated "*a more positive attitude towards those who work remotely*" and that NTCs would be used more as "*a natural place where you can get away from home.*" Most of the

participants believed that the pandemic would have a long-term impact on views of remote working:

> *"Sceptical managers will be able to let go of all that now. Everyone knows it works."*
> *"We had a tough policy. Now everyone is home and people are talking about getting everything done and having time left over. The boss kind of says, 'We will have to re-evaluate [remote work]'.*

Some thought that the pandemic had proven that technology can be used to improve remote working:

> *"The technology has been around for a long time. /…/ Those who were unfamiliar have now learned. Employers need to rethink; they are noticing that it works with the new technology."*

Another participant, who had been a regular user of the NTC, expressed that:

> *"It's easier when everyone works remotely. Everyone is equal in the meetings."*

Several participants made it clear that if their workplace continued to allow more remote working, they would like to be at the NTC more often:

> *"[Normally] I very rarely work from home. I have better focus elsewhere. Yes, I will definitely use the [NTC]* (after the pandemic)*. It's nice too, you get to know some different people."*
> *"The [NTC] will be much more important. We can work remotely, but we also have to make use of our experiences of [COVID-19], that we need the social bit."*

5. Discussion

As seen in Fig. 2 and presented in the interview results in section 3.1, it cannot clearly be said that the NTC prototype in the DLL *Work Closer, Travel Smarter* led to a direct reduction in travel. The main reason for this was that most participants could not work remotely more than one day a week, and therefore mostly exchanged working from home for working at the NTC. For these people, the NTC added a short trip from home to the NTC, and thus had a reverse effect. However, there are some indications that the NTC may still hold potential for enabling people to avoid travelling by reducing their need for commuting.

Firstly, for the few participants who used the NTC frequently, it did lead to shorter trips, as they exchanged their normal train commute for a walk to the NTC. Secondly, some participants started shopping locally on their way home from the NTC rather than shopping separately on the weekend. In the short term, this meant that they potentially replaced longer shopping trips with shorter ones on their way home from work. At scale, this could contribute to keeping smaller city centres open. Thirdly, for one of the participants, using the NTC motivated her to find a new job closer to home.

As most participants normally travelled to work by commuter train, and the NTC was placed next to the commuter train station, the travel mode that the participants would normally take to the train station, often bus, bicycle or walking, now became the main travel mode. In some cases, however, participants replaced travelling to the employer's office by train, with going to the NTC by car. There were indications that many participants commuted by train mainly due to congestion, but preferred the car whenever this was not an issue, which is in line with previous findings by Sjöman et al. (2020). Further, when working closer to home, the car was sometimes considered more convenient for combining personal and work activities. Some of the participants reported shopping more locally in connection with using the NTC. While we do not know whether this amounted to avoiding a separate car trip, such changes would be interesting to explore further from a land use and urban planning perspective. That several of the participants bought their own e-bicycles in connection with working more at the NTC points to some potential to support bicycle ridership. Additionally, many of the participants expressed that the commuter train ran at full capacity during peak hours, which was their normal commuting time. In this case, NTCs may help free up public transport, but in places where public transport ridership is low, they may risk further undermining it.

Originally, the aim of this study was to present the NTC as an alternative to travel options on a digital platform, as has been suggested in the MaaS-literature (Jittrapirom et al., 2017). However, none of the participants used the digital platform as intended. This can be partly attributed to problems with the functionality and usability of the booking app, which received a lot of complaints. However, it was also clear that the participants had established regular routines around commuting, and would just walk to the station to take the train as usual. The majority also owned commuter cards and did not need to purchase trips through the platform. This corresponds to the findings of Sjöman et al. (2018) that commuting is likely not a very interesting case for MaaS. Also, the participants saw no use for any additional services, as they used the private car for most of their leisure time travel.

On the question of whether remote working contributes to urban sprawl by encouraging living far from one's workplace, we saw no indications of this in our results. On the contrary, one of the participants changed workplaces to be able to work closer to home. Furthermore, many participants had settled in the area when their workplace was nearby and stayed there even as the workplace moved.

5.1. Future sustainability potentials of NTCs

There are several conditions that affect the possibilities for NTCs to support more sustainable travel.

5.1.1. Placement

Firstly, the placement of the NTC next to the station was suggested by several of the participants to be convenient, both for those who travelled one or two stops by commuter train to get to the NTC and for those who liked to sit at the NTC for part of the day, often on their way home from the commuter train. In that sense, the placement next to the station likely encouraged travelling to the NTC by train. The possibility to easily stop by at the NTC on the way to or from the office could also help to even out the load on the commuter train. On the other hand, although the NTC was placed in a neighbourhood with high car ownership, it mainly attracted workers who normally commuted by train. In that sense, it rarely resulted in modal shifts away from the car. An NTC would likely be more effective in an area with a larger share of car commuters, and less access to public transport.

5.1.2. Practicalities

Secondly, many participants shared discontent with the functionality of the living lab NTC prototype. In most cases, this could be easily remedied in the future, with better phone/video meeting booths and more computer monitors. For people in roles that involve many meetings, a private room is a strong requirement. An NTC with practicalities set to match workers' needs for privacy would likely attract more users. It may, however, be difficult to provide a suitable environment within the framework of an NTC.

5.1.3. Policies

Thirdly, some workplace policies and roles strictly required being present at the employer's office, such as project management and educating new colleagues. Overall, the need to be in place for collaboration and cooperation was mentioned by many of the participants as the main reason for not being able to avoid the trip to the office. Sometimes, this had to do with in-person meetings where the participant would be one of few who called in if they were at the NTC. During COVID-19, when all or most meetings were attended virtually, this was no longer seen as a problem. Most participants envision policy changes towards allowing more remote work in the future. An NTC used by

employers who have policies allowing employees to work remotely would likely have a greater influence on reducing the need for commuting.

In this study avoiding travel meant replacing it with a non-transport related alternative in the shape of an NTC. In line with this, observed variations in travel behaviour change have mostly been related to work; the office layout of the NTC prototype, work-life regulations and organizational aspects. This deserves a special note, as, on these lines, we believe that research into means for avoiding transport will often have to include more interdisciplinary perspectives, as it will need to place more focus on societal dimensions.

6. Conclusions

Overall, remote work reduces the need for work-related travel, but the effects of gaining access to an NTC were not clear. As most participants could not be away from the office more than 2–4 days per month, they merely exchanged working from home for working from the NTC. For a small number of participants who lived in close proximity to the NTC and whose working conditions allowed using it often, the NTC triggered more significant lifestyle changes. These participants walked to the NTC, used the e-bicycles to make trips during the day, and ran errands more locally, amounting to positive change. Several others expressed a desire to work more at the NTC and avoid the long commute, but felt that this was not possible. We identified several conditions that made it difficult for most participants to avoid going to the office: mainly workplace regulations, norms and expectations, and work tasks and roles. With the onset of the COVID-19 pandemic, the participants observed changes in attitudes towards remote work, effectively eradicating many of these identified barriers for remote work. Many participants expressed the need for NTCs during this period, believing that they would use one more often after the pandemic to remedy some of the negative aspects of working from home. While it remains unclear how many people would need or choose an NTC in the future, attitudes towards remote working seem to have changed dramatically since the start of the pandemic. If these changes persist, NTCs could play a role in enabling more remote working in the long term, potentially supporting more local and sustainable lifestyles, and reducing the need for travel.

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